

The Normal Distribution: $N(\mu, \sigma^2)$

A random variable x is said to have a normal distribution with mean μ ($-\infty < \mu < \infty$) and variance $\sigma^2 > 0$ if it has the density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty.$$

Notation: $x \sim N(\mu, \sigma^2)$ to indicate that the random variable x is normally distributed with mean μ and variance σ^2

Properties:

(i) $\int_{-\infty}^{\infty} f(x) dx = 1$ ✓ required for all density functions

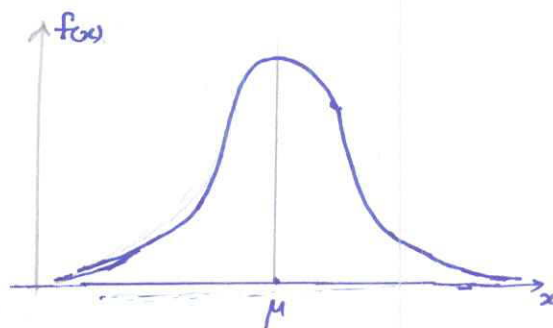
(ii) $f(x) \geq 0$ for all x

(iii) $\lim_{x \rightarrow \infty} f(x) = 0$ & $\lim_{x \rightarrow -\infty} f(x) = 0$

(iv) $f(\mu+x) = f(\mu-x)$. The density is symmetric about μ .

(v) The maximum value of f occurs at $x = \mu$.

(vi) The points of inflection of f are at $x = \mu \pm \sigma$.



The moment generating function for the normal distribution can be shown to be

$$M_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

The mean and variance of the normal distribution is $E(x) = \mu$ and $V(x) = \sigma^2$, respectively. ✓ ✓

The Normal Cumulative Distribution Function:

The distribution function F is $F(x) = P(X \leq x) = \int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{u-\mu}{\sigma}\right)^2} du$.

It is impossible to evaluate this integral without resorting to numerical methods, and even then the evaluation would have to be accomplished for each pair (μ, σ^2) . However, a simple transformation of variables, $z = (x - \mu)/\sigma$, allows the evaluation to be independent of μ and σ . That is,

$$\begin{aligned} F(x) &= P(X \leq x) = P\left(Z \leq \frac{x-\mu}{\sigma}\right) = \int_{-\infty}^{(x-\mu)/\sigma} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \\ &= \int_{-\infty}^{(x-\mu)/\sigma} \phi(z) dz = \Phi\left(\frac{x-\mu}{\sigma}\right). \end{aligned}$$

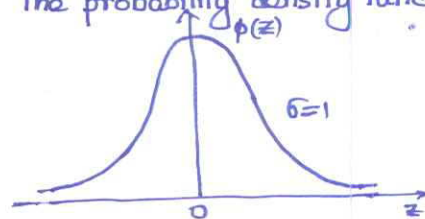
The Standard Normal Distribution: $N(0, 1)$

If $Z \sim N(0, 1)$, then Z has a standard normal distribution. The probability density function of the standard normal density function is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty.$$

The corresponding distribution function is Φ , where

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$



Ex: The breaking strength (in newtons) of a synthetic fabric is denoted X and it is distributed as $N(800, 144)$. The purchaser of the fabric requires the fabric to have a strength of at least 772 nt. A fabric sample is randomly selected and tested. Find $P(X \geq 772)$.

$$P(X < 772) = P\left(\frac{X - \mu}{\sigma} < \frac{772 - 800}{12}\right)$$

$$= P(Z < -2.33)$$

$$= \Phi(-2.33) = 0.01$$

Hence the desired probability, $P(X \geq 772)$, equals 0.99.

$$P(X \geq 772) = 1 - P(X < 772)$$

$$= 1 - P(X - \mu < 772 - \mu)$$

$$= 1 - P\left(\frac{X - \mu}{\sigma} < \frac{772 - \mu}{\sigma}\right)$$

$$= 1 - P(Z < -2.33)$$

Ex. The pitch diameter of the thread of a fitting is normally distributed with a mean of $\mu = 0.4008$ cm and a standard deviation of $\sigma = 0.0004$ cm. The design specifications are 0.4000 ± 0.0010 cm. Notice that the process is operating with the mean not equal to the nominal specifications. We desire to determine what fraction of product is within tolerance.

$$P(0.399 \leq X \leq 0.401) = P\left(\frac{0.3990 - 0.4008}{0.0004} \leq \frac{X - \mu}{\sigma} \leq \frac{0.4010 - 0.4008}{0.0004}\right)$$

$$= P(-4.5 \leq Z \leq 0.5)$$

$$= P(Z \leq 0.5) - P(Z \leq -4.5)$$

$$= \Phi(0.5) - \Phi(-4.5)$$

$$= 0.6915 - 0$$

$$= 0.6915$$

- As process engineers study the results of such calculations, they decide to replace a worn cutting tool and adjust the machine producing the fittings so that the new mean falls directly at the nominal value of 0.4000. Then,

$$P(0.3990 \leq X \leq 0.4010) = P\left(\frac{0.3990 - 0.4}{0.0004} \leq Z \leq \frac{0.4010 - 0.4}{0.0004}\right)$$

$$= P(-2.5 \leq Z \leq +2.5)$$

$$= \Phi(2.5) - \Phi(-2.5)$$

$$= \Phi(2.5) - (1 - \Phi(2.5))$$

$$= 2\Phi(2.5) - 1$$

$$= 2(0.9938) - 1$$

$$= 0.9876$$

$\therefore$ We see that with the adjustments, 98.76% of the fittings will be within tolerance.

- Note: About 68% of the area will be covered by the interval $\mu \pm \sigma$
 95% area will be covered by the interval $\mu \pm 2\sigma$
 99.7% area will be covered by the interval $\mu \pm 3\sigma$

$$P(\mu - \sigma < X < \mu + \sigma) = P(-\sigma < X - \mu < \sigma)$$

$$= P(-1 < \frac{X - \mu}{\sigma} < 1)$$

$$= P(-1 < Z < 1)$$

$$= \Phi(1) - \Phi(-1)$$

$$= \Phi(1) - (1 - \Phi(1))$$

$$= 2\Phi(1) - 1$$

